

1. Deuterium ions have twice the mass of a proton but carry the same charge. What would be the radius of curvature of magnetically deflected deuterium ions if the magnetic field strength were $4.0 \times 10^3 \text{ T}$ and the accelerating voltage were $1.0 \times 10^3 \text{ V}$?
2. A magnetic field from a solenoid is used to deflect a beam of electrons by **0.8 cm** on the screen of a CRT. If the current in the solenoid is doubled, what will the deflection be then?
3. An electron beam in a CRT is deflected **0.64 cm** when the accelerating voltage is **500 V**. By how much will the beam be deflected if the magnetic field is kept constant but the accelerating voltage is increased to **2000 V**?

4. An alpha particle is accelerated by a voltage of $1.53 \times 10^3 \text{ V}$ and is then deflected by a magnetic field of strength 0.020 T into a circular path of radius 0.40 m . If the alpha particle has a charge of $3.2 \times 10^{-19} \text{ C}$, what is its mass?
5. In the velocity selector part of a mass spectrometer, a magnetic field of 0.65 T is used for magnetic deflection of a beam of protons having a speed of $1.0 \times 10^6 \text{ m/s}$. What electric field is needed to balance the force due to the magnetic field? If the distance between the plates of the electric deflection apparatus is 0.50 cm , what voltage must be applied to the plates?
6. What speed must the electrons in a beam passing through a velocity selector have if the beam is undeflected by crossed electric and magnetic fields of strength $6.0 \times 10^3 \text{ V/m}$ and 0.0030 T , respectively? If the electric field is shut off, what would be the radius of the beam due to the unbalanced magnetic force?

